

Clustering in A Sensor Array System Based on The Distribution of Volatile Compounds from Palm Oil Using Electronic Nose

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Abstract—Unsaturated fat in palm cooking oil becomes dangerous if repeated heating to become trans-fat which is carcinogenic. Many circulating used cooking oils that are not fit for consumption are clarified in various ways, but this process does not change the composition of the harmful substances it contains so that it remains dangerous if consumed. In previous studies, the method often used was Gas Chromatography - Mass Spectrometry (GC-MS), by providing complete compound results. However, the use of this method has a long process and high costs. The previous method e-nose (electronic nose) was used to differentiate cooking oil from a mixture of olive oil and lard. However, there has been no research that focuses on e-nose to detect the quality of cooking oil. In this research, the e-nose was used to differentiate between used cooking oil and fresh cooking oil. Based on the clustering results, there is a distribution that separates fresh and used oil. The results of this study are expected to be an alternative to measure the quality of cooking oil quickly and easily. In addition, the results of this study can also assist in efforts to improve public health through food quality control.

Keywords—*Electronic Nose, Cooking Oil Quality, Clustering, Fresh Cooking Oil, Signal processing*

I. INTRODUCTION

Consumption of cooking oil is very large in all circles. Used oil or oil that has been heated excessively so that palm oil will contains saturated fat and unsaturated fat when heated it will oxidize and hydrolyze resulting in the formation of trans fats. Unsaturated fat that is heated above 200°C can turn into trans-fat [1]. Chemical structure of saturated fat, unsaturated fat and trans-fat. Consumption of trans fat will increase Low Density Lipid (LDL) and reduce High Density Lipid (HDL) thereby causing cardiovascular disease[2]; high incidence of genotoxic, mutagenic, tumorigenic, and various types of cancer[3]. Trans fat is carcinogenic which causes breast cancer, shortens gestational age, risks of preeclampsia, disorders of the nervous system and vision in infants, colon cancer, diabetes, obesity and allergies. WHO data for 2015 shows that 1.13 billion people have hypertension, which means that 1 in 3 people has hypertension. Many used clarified

oils are sold in the market [4] this is reasonable because oil quality testing is rarely carried out on sellers and buyers in small shops.

Cooking oil quality testing generally uses GC-MS (Gas Chromatography-Mass Spectrometry), which is expensive and takes a relatively long time, so a cheap and fast cooking oil quality detector is needed. In addition, a tool is needed that can be used anywhere so that it can be used directly in the field. Data from the Large and Medium Industries (IBS) for 2021 shows that there are 74 palm cooking oil factories across Indonesia and 45 factories on the island of Java. Apart from being on an industrial scale, tools are needed that can be used by consumers, including shops and buyers, so that the potential for the tools to be developed can be used by all groups. In the context of cooking oil quality, GC-MS can be used to identify and measure components such as free fatty acids, peroxides, aldehydes, ketones and other compounds formed during oil processing and storage [5]. GC-MS analysis can provide information on oil degradation, oxidation, thermal instability and general quality[6].

This research develops the e-nose tool to detect the quality of cooking oil. The working principle of e-nose mimics the nose's sense of smell [7] and has been applied to detect Covid-19 disease and respiratory tract infections[8], pork mixture in food [9], [10], civet coffee[6], diabetes level [11], pineapple sweetness level [12], to predicting the oxidative stability of biodiesel in oil, detecting differences in types of oil, adulteration of cooking oil mixtures, classification and quality control of oil and a mixture of olive oil and lard. The advantages of e-nose are high sensitivity, fast, and ability to analyze small samples [13]. e-nose is a device consisting of several chemical sensors that are used together with software for pattern recognition. This tool generates semi-quantitative data resulting from sensor readings that are regressed from GC-MS data, so that e-nose can determine a substance by recognizing the gas of that compound [14], [15], [16]. Previous studies have not focused on detecting the quality of cooking oil using e-nose, so the results of this study are

expected to be an alternative for measuring the quality of cooking oil quickly, easily and accurately.

Additionally, the outputs from the e-nose will also be processed using clustering algorithms. Clustering is an unsupervised learning method capable of grouping data points based on inherent similarities, thereby revealing underlying patterns within complex datasets. By applying clustering algorithms, the process of predictive modeling to detect cooking oil quality can be easily facilitated [15].

II. RESEARCH METHOD

A. Quality of Cooking Oil

The quality of cooking oil can be affected by several factors which include oxidation, formation of degradation compounds, thermal stability, and physicochemical changes. The following is a brief explanation of each of these factors:

- Oxidation of Cooking Oil:** Oxidation is a chemical reaction that occurs between cooking oil and oxygen in the air. Oxidation can lead to the formation of oxidative compounds and free radicals, which can change the taste, aroma, color and nutritional value of cooking oil. Free fatty acid levels and peroxide numbers are often used as indicators of oxidation in cooking oil [7].
- Degradation Compounds:** Repeated heating of cooking oil can cause the formation of degradative compounds such as free fatty acids, aldehydes, ketones and polar compounds. These compounds can affect the taste, aroma and nutritional quality of cooking oil [2].
- Thermal Stability:** Thermal stability refers to the ability of cooking oil to maintain its stability when heated to high temperatures.

B. Gas Sensor

Gas sensors are electronic devices that detect various gases in the air and identify signals in accordance with the parameters [17]. Metal Oxide Semiconductor (MOS) sensors are among the most widely used gas sensors. This sort of sensor includes a chemiresistor that interacts with the target gas [18]. When MOS sensors come into touch with gases like carbon monoxide, hydrogen, methane, or butane, their electrical resistance increases. The sensor consists of two sections, one of which is the detecting membrane, which reacts or detects the target gas. Fig. 1 illustrates how the transducer translates the presence of gas into a signal [19].

In this study, 8 sensors with the MOS type were used, each of which was initialized as S1, S2, S3, S4, S5, S6, S7, S8. The MOS gas sensor is then controlled using a microcomputer that has been programmed to run according to the needs of data collection to then be sent to the computer and then pre-processing and learning processes are carried out.

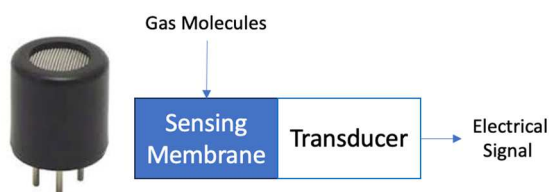


Fig. 1. Gas Sensor

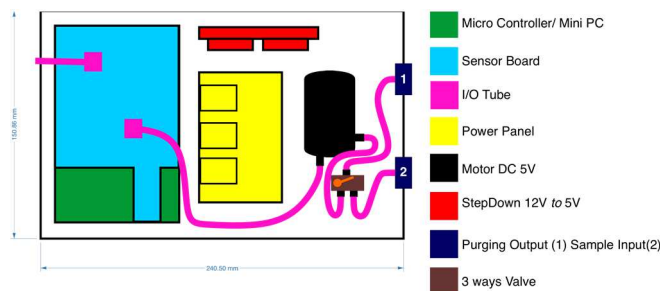


Fig. 2. Electronic Nose

C. Electronic nose

The electronic nose is a tool specifically designed to mimic the workings of the human sense of smell. E-nose can identify an object based on its smell and based on the chemicals contained in it or Volatile Organic Compound (VOC). Based on Fig. 2, an e-nose device consists of several components, namely a vacuum pump for taking samples, a chamber, a sensor array chamber, a power supply, a control circuit, a fan, a voltage divider, a headspace system, and signal data converter from analog to digital (ADC)[20].

The e-nose in this study used 8 sensors from the TGS and MQ series available in Indonesia with an additional temperature and humidity sensor. Each sensor has sensitivity to its own chemical substances. Then for the Data Acquisition System (DAQ) using a Raspberry mini-computer [21], [22], [23], [24]. The output of e-nose is in the form of digital data which will be processed using a computer using the Python programming language and several libraries.

III. PROPOSED METHOD

A. E-nose Hardware

E-nose hardware systems consist of various hardware components that enable the detection and analysis of odors and volatile compounds. The core component of an e-nose is the sensor array, which consists of multiple chemical sensors with different selectivity and sensitivity to volatile compounds. Each sensor in the array has a specific selectivity and sensitivity to certain odor molecules, allowing for broad odor detection and differentiation. In this case the sensors used are 8 types of sensors with each sensitivity. And the other component is sample chamber, where the odor or gas sample is introduced for analysis. It provides a controlled environment for the sensors to interact with the volatile compounds present in the sample. The chamber has a driving component in the form of a vacuum pump to push the sample which is pulled from the sampling glass. In the sampling glass provided, the cooking oil to be tested will be put into it with a composition of 150 ml for each sample at the same temperature and room conditions.

In simple terms, the e-nose can be described as shown in Fig.2. An e-nose has a sensor box or sensor chamber, a vacuum pump, a Data Acquisition System (DAQ) that uses a micro controller, and other supporting components with a 12volt DC input.

B. Sample Preparation

Collection was carried out independently in the laboratory with various samples of new cooking oil and used oil. The

used oil is oil of the same brand but heated to a temperature of 200 degrees for 5 minutes without any mixture of ingredients when frying to avoid odors from other objects. The following are the steps in preparing for sampling:

- Identification and labeling: put each label in a beaker glass containing 150 ml of cooking oil.
- Sample heating: Sampling was carried out by heating to 80 degrees Celsius using an electric stove with automatic temperature control
- Sampling: Sampling was carried out with e-nose by inhaling gas from cooking oil steam after heating to 80 degrees Celsius.

This step was carried out until all samples were taken as a whole. All samples of new and used cooking oil must be cold, or no more than 80 degrees Celsius when tested.

C. Sample Process

E-nose can take measurements using cooking oil samples with several predetermined procedures. The e-nose system design can be seen in Fig. 3.

Stages of sample testing using an electronic nose will be carried out in several stages. First, the preparation stage, cooking oil samples were taken and prepared using a beaker glass with a total of 150 ml and covered with a lid connected to an oil heater, because if the temperature is more than that, the oil will smoke and make the e-nose have a very high value, and if it is lower than that, the value is too small and even has the same value as the base value. The sample acquisition system design can be seen in Fig. 4. Then at the measurement stage, the cooking oil sample will detect the aroma pattern and produce an output value that represents the aroma characteristics of the sample.

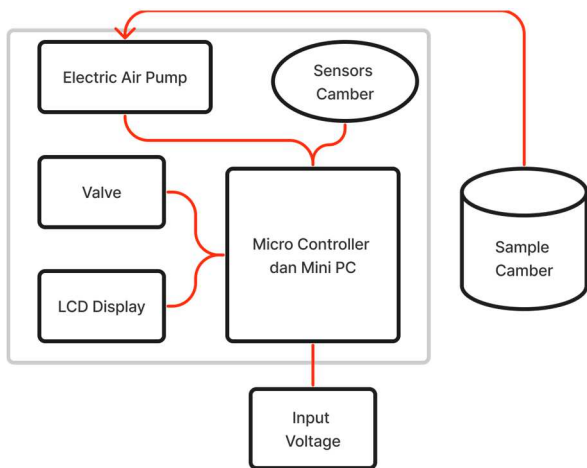


Fig. 3. Sampling Design

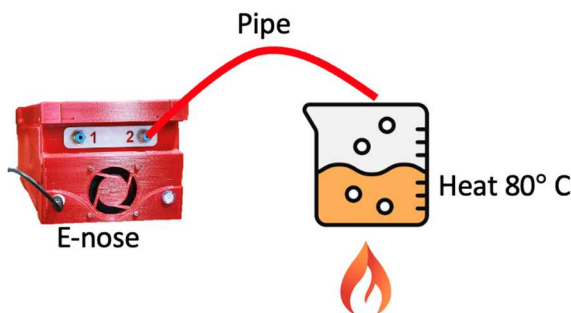


Fig. 4. Sample Acquisition

IV. EXPERIMENTAL RESULT

A. E-nose Sensor Reading Results

The e-nose sensor reading is done by measuring the sensor's response to the aroma or gas present in the sample. Each sensor on the e-nose has unique characteristics and sensitivity to certain compounds. During reading, the sample is exposed to the first 8 sensors, and the sensor response is converted into an electrical signal which is then processed and analyzed.

This research applies e-nose which consists of several sensors that are sensitive to volatile compounds related to the quality of cooking oil. Each sensor produces a different response when exposed to the aroma from the cooking oil sample. This response is a unique signal pattern for each type of oil, both fresh and used. This signal pattern reflects the chemical composition of the volatile compounds in the oil.

The process of reading the e-nose sensor involves absorbing the scent by the sensor, converting the signal into digital form, and analyzing the pattern of the resulting signal. the results obtained are graphed and compared by looking at the signal patterns of the samples. Previously known signal patterns are used to identify and classify the quality of cooking oil. Thus, the reading of the e-nose sensor provides information about the aroma or gas profile in the cooking oil sample which is related to its quality. The results of these readings can be used to identify and differentiate fresh oil from used cooking oil, as well as provide information about the purity and condition of the oil.

Note that you need to know that every time you take e-nose data, always pay attention to the room conditions when taking samples. the room must be in a condition that does not have a strong odor such as room freshener or something sprayed on it. some readings may be different for each room so you have to pay attention to the neutral position of the sensor readings whether they are still in a position that does not exceed the limit

B. Data Processing

The e-nose signal consists of 3 parts as shown in Fig. 5 namely the Base Line (P1) which originates from free air as a neutral position, then the Data Line (P2) which is the sampling process, and the Purging Line (P3) is the data cleaning process for remove the odor from the remaining sample residue on the camber e-nose. The x-axis shows the seconds, and the y-axis shows the voltage generated by the e-nose in millivolt form as Fig. 5-7.

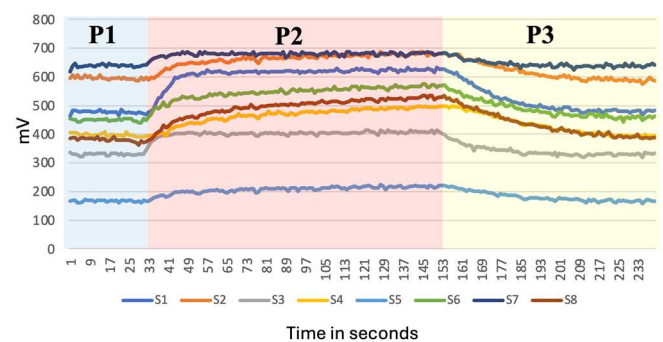


Fig. 5. Part of the sample

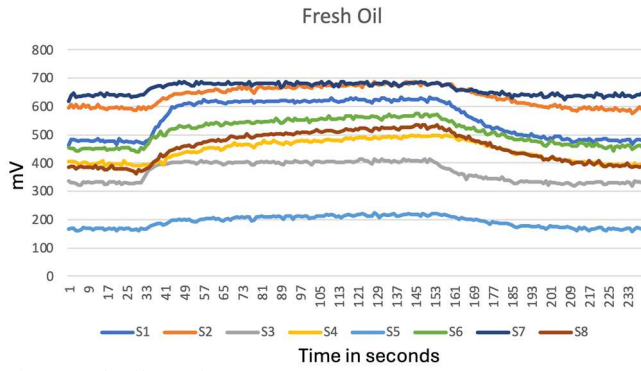


Fig. 6. Fresh Oil Samples

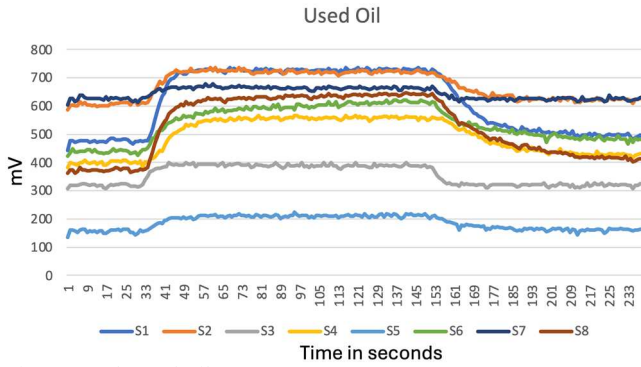


Fig. 7. Sample Used Oil

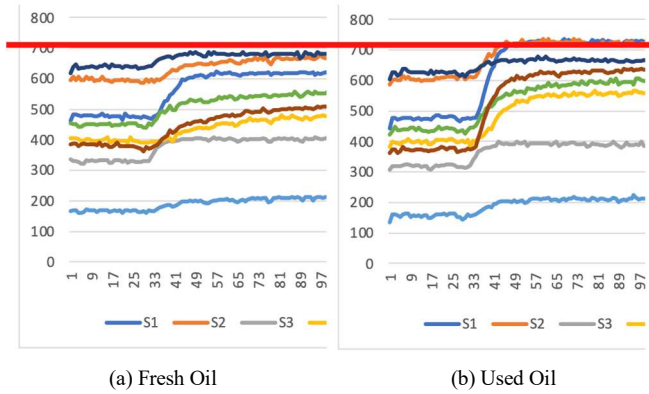


Fig. 8. Comparison of Fresh Oil with Used Oil

From Fig. 8 it can be seen that the results of the e-nose readings show differences that can be observed from the movement of each graph. Observed from P2, it shows that fresh oil looks more sloping than used oil which has a higher rise. Consecutive sensor readings from S1 to S8 can be calculated using the mean from P2 but taken when the position is steady, namely at the 50th to 140th second, each of these values is from S1-S8 for 619mV fresh oil; 668mV; 402mV; 474mV; 210mV; 551mV; 680mV; 505mV and S1-S8 for used oil is 726mV; 720mV; 388mV; 557mV; 211mV; 598mV; 663mV; 623mV. In this case, almost all sensors look different except for S3, S4 and S7, which have no significant increase, even for S3 and S7, they experience a slight decrease. Meanwhile, the highest increase was obtained in S1 and S2.

Overall, the findings highlight the effectiveness of the e-nose system in detecting and differentiating between new and used cooking oil based on their aroma. However, it should be noted that although the e-nose can distinguish various types of cooking oil, it cannot provide specific information about the composition of the compounds detected in the oil, so it is

necessary to do a comparison with laboratory equipment to show specific differences in the content of new or used oil.

C. Result of Clustering

In this study, there are a total of 14 oil samples tested, labeled name are marked with [Oil brand][time in hour] such as SA00, SA10, SA20, SA30, SA40, SA50, SA60, SIP00, SIP10, SIP20, SIP30, SIP40, SIP50, and SIP60. The codes on the sample labels have specific meanings, where brand with name 'SA' indicates expensive oil in market, while the number next to the letter code represents the duration of heating. Descriptions related to the sample labels will be explained in TABLE I.

Each sample was tested repeatedly 5 times to ensure sensor stability, resulting in a total of 140 sensor reading data. 70 data points were taken in December 2023 and another 70 data points were taken in January 2024.

Next, the Clustering process was applied to 35 cheap oil data to determine whether the e-nose can detect whether an oil is new or has been used and for how long it has been heated. The clustered data consists of 5 samples each of SIP00, SIP10, SIP20, SIP30, SIP40, SIP50, and SIP60. This study uses the Python programming language and utilizes the scatter plot library for the clustering process. Fig. 9 shows the clustering results of the cheap oil data.

TABLE I. DESCRIPTION OF LABELS ON THE SAMPLE

Label	Description
SA00	Expensive oil that has never been heated.
SA10	Expensive oil that has been heated for 10 hours.
SA20	Expensive oil that has been heated for 20 hours.
SA30	Expensive oil that has been heated for 30 hours.
SA40	Expensive oil that has been heated for 40 hours.
SA50	Expensive oil that has been heated for 50 hours.
SA60	Expensive oil that has been heated for 60 hours.
SIP00	Cheap oil that has never been heated.
SIP10	Cheap oil that has been heated for 10 hours.
SIP20	Cheap oil that has been heated for 20 hours.
SIP30	Cheap oil that has been heated for 30 hours.
SIP40	Cheap oil that has been heated for 40 hours.
SIP50	Cheap oil that has been heated for 50 hours.
SIP60	Cheap oil that has been heated for 60 hours.

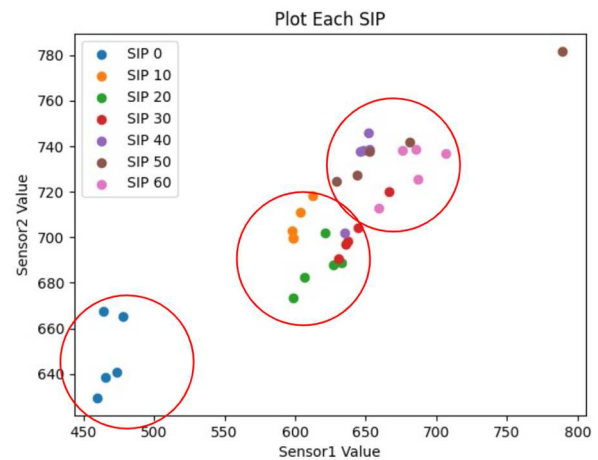


Fig. 9. Clustering Results of the Cheap Oil data.

From Fig. 9, it can be seen that broadly, the distribution of e-nose reading data is divided into 3 major parts, which are then labeled as Class Oil A, Class Oil B, and Class C to facilitate the analysis process. Oil A is the category of oil that has never been heated, Oil B is oil that has been heated for 10-30 hours, while the Oil C category is oil that has been heated for 40-60 hours. This experiment proves that the e-nose can identify how long an oil has been heated.

V. CONCLUSION

Based on the results of the study, it can be concluded that the e-nose technology is able to distinguish between new and used cooking oil well. The use of appropriate data processing methods and accurate modeling allows e-nose to distinguish between new and used cooking oil samples very well. E-nose provides clear and reliable results in identifying differences in aroma and odor characteristics between new and used cooking oil. E-nose can be an alternative in providing fast measurements or used as an initial identification in determining the quality of cooking oil.

However, it should be noted that the e-nose has some drawbacks. One of them is that e-nose is unable to provide specific information regarding the content of compounds present in cooking oil. Although e-nose can distinguish between new and used oil based on the pattern of aroma or odor produced, it does not provide detailed information about the specific compounds contained in the cooking oil. Therefore, the use of additional analytical techniques such as GC-MS (Gas Chromatography-Mass Spectrometry) is still required to identify.

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